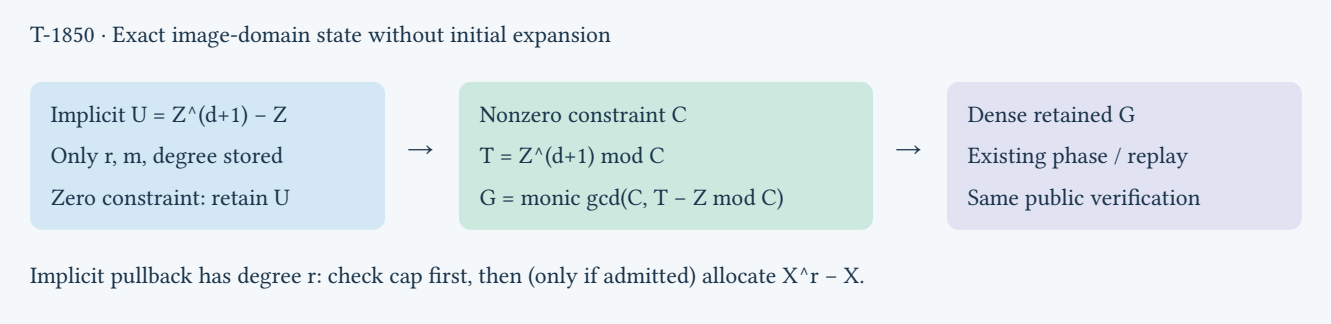
Keep the initial image domain implicit

T-1850 · 2026-11-07 · Exact scalar-field recovery representation

Removed the unconditional dense initialization of U(Z)=Z^(d+1)−Z, d=(r−1)/m, from the traced recovery path. The initial state now holds only parameters until a nonzero constraint arrives. This targets an allocation before observed constraints and before the fiber cap, independently of the source quotient trace.



Proof. Gcd(U,C)=gcd(C,U mod C). Binary modular powering supplies the required residue without expanding U. Zero constraints retain implicit state; nonzero constants produce the unit polynomial. Subsequent dense intersections are unchanged. Induction gives the same state after every admitted constraint sequence, including repeated or nonmonic constraints.

Cap and costs. Implicit pullback is U(X^m)/X^(m−1)=X^r−X. Reject degree r greater than cap before allocating. First intersection with degree-c constraint uses residues below c and multiplication outputs at most degree 2c−2. Powering costs floor(log2 N) squarings plus popcount(N) multiplications/reductions, N=d+1, then gcd. This trades modular-power work for avoiding the full-domain array; no universal timing dominance is claimed. For m≤(r−1)/2 the first mode constraint has degree floor(d/2), with nonzero leading coefficient.

Deterministic native check	Count	Result
Constraint intersections	8,600	Exact dense-gcd and pullback agreement
Zero stages remaining implicit	2,192	No dense state created
Cap rejection controls	3,113	Limit returned
Large polynomial helper	N=2^60	No U expansion; exact linear intersection
Additional public recoveries	617	5,929 calls; zero unresolved
Inherited public comparisons	63	511 calls; same results/routes

Scope. The large helper uses prime r=2^61−1 and C=Z−1; it is not a large-order public-source recovery. Mode coefficient construction and the coordinate-field source grid remain separate costs. Runtime target access remains public-point-only. Original 4,563-case cohort and the prospective trace protocol are unchanged. No cloud, timing or full Cargo rerun.

Evidence and next construction. DERIVATION.md supplies invariants, costs, assumptions and replay command; native receipt pins dependencies and output. PCC-20261107-83d4e9 is unsigned producer evidence. Sources: exact predecessor source/recovery/gcd and FLINT modular exponentiation; KB unavailable. Next develop a compact or streamed first-constraint representation, since the initial domain now intersects at that constraint's size. Historical evidence and canonical research graphs remain unchanged.